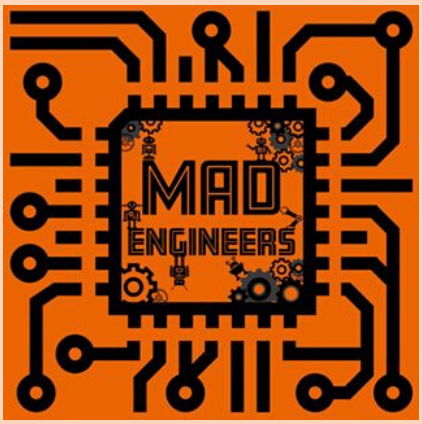
FTC TEAM #2123 MAD ENGINEERS

WIRING AND AUTONOMOUS FUNCTIONS DOCUMENT



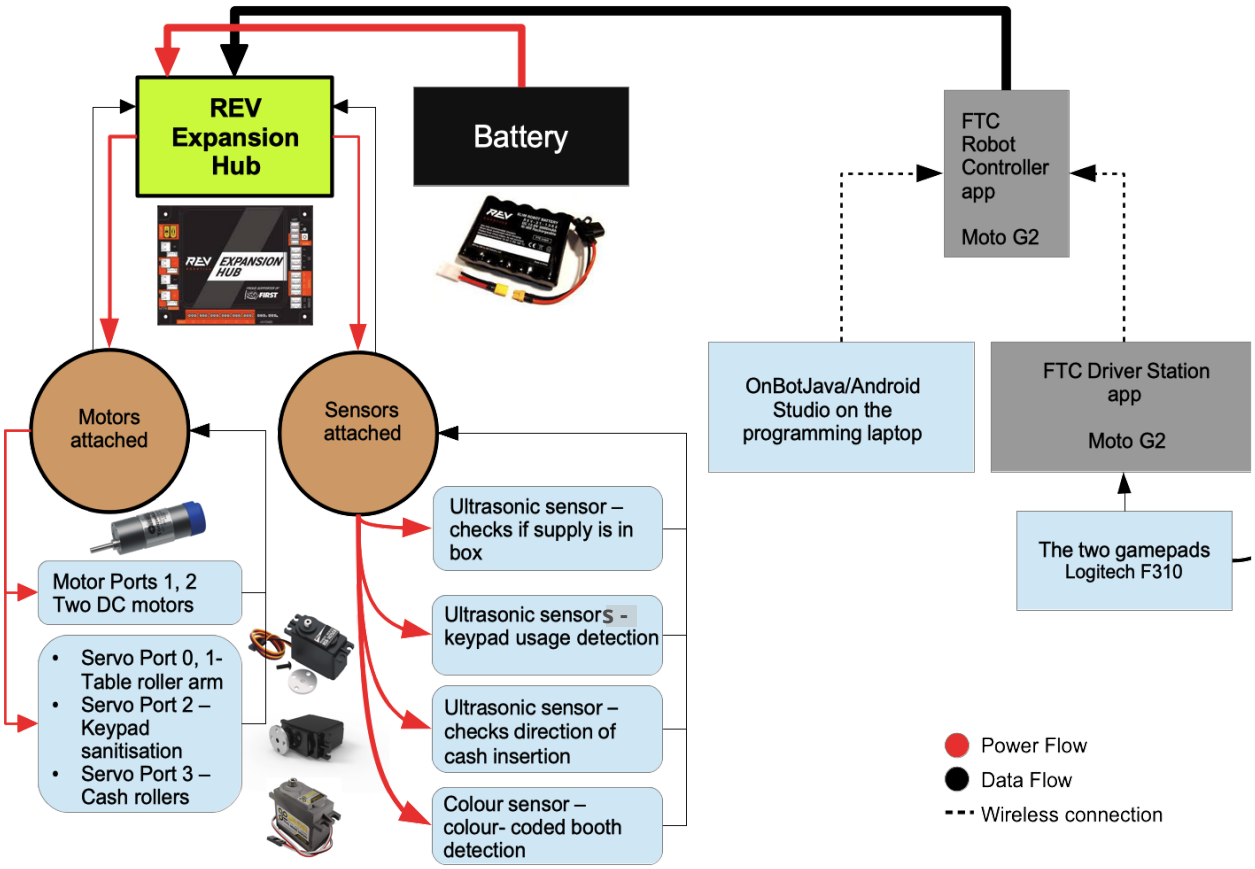
LOCATION Mumbai, India

SCHOOL On My Own Technology

FIRST TECH CHALLENGE

Onshape

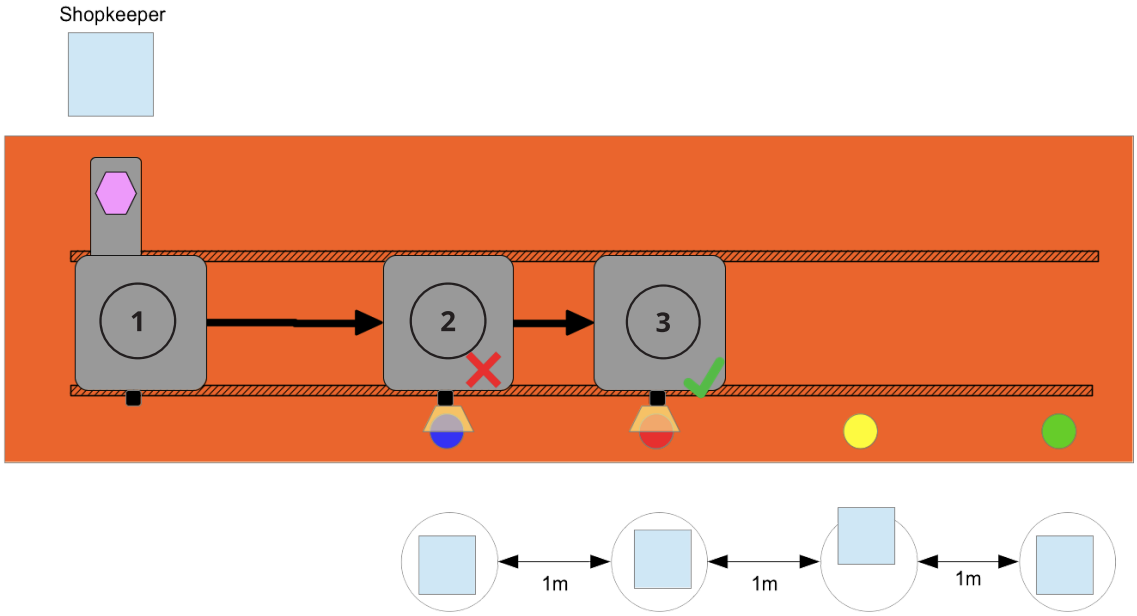
Wiring schematic



This diagram represents the data and power flows in our robot.
All DC and servo motors, digital and analog sensors attached to our robot are represented in this diagram.

Supply Delivery Mechanism - PART 1

Note: the robot will be placed over-the-counter, on pre-defined tracks. The below is a top view of a counter in a typical shop setting. The black lines represent the tracks.

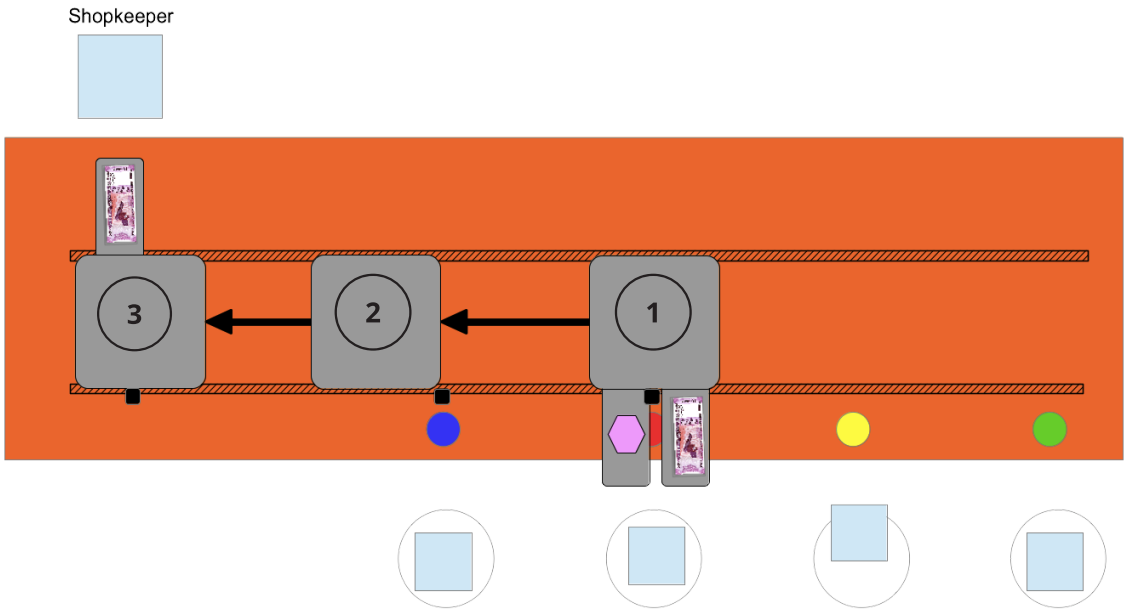


POSITION 1:
In the starting position, the shopkeeper places all the goods in the supplies box, where they are automatically packaged. The robot then starts moving towards the customers.

POSITION 2:
Colour sensor detects blue. However, since the package is for the second booth, the robot keeps moving till it detects the colour red.

POSITION 3:
The robot performs Vuforia object detection to ensure that a customer is present. If this is true, it stops its motion. The customer can now pick up the packaged supply.

Supply Delivery Mechanism - PART 2



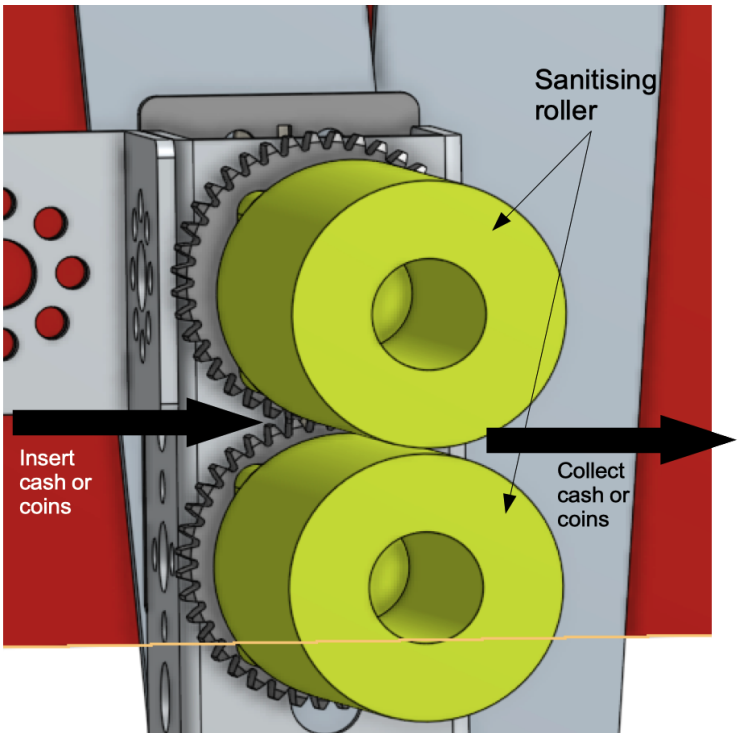
POSITION 1:
After picking up the supply, the customer can pay through the cash or coin by inserting it through the rollers, or by using the installed credit card machine. The bill generated from the billing machine can be collected.

POSITION 2:
While moving from Position 1 to Position 2, the cash or coins pass through the roller and get sanitised. After this is done, they fall into the collection tray on the side of the robot.

POSITION 3:
Once the robot reaches the shopkeeper, he can pick up the currency. There is the option to return the robot to provide change to the customer.

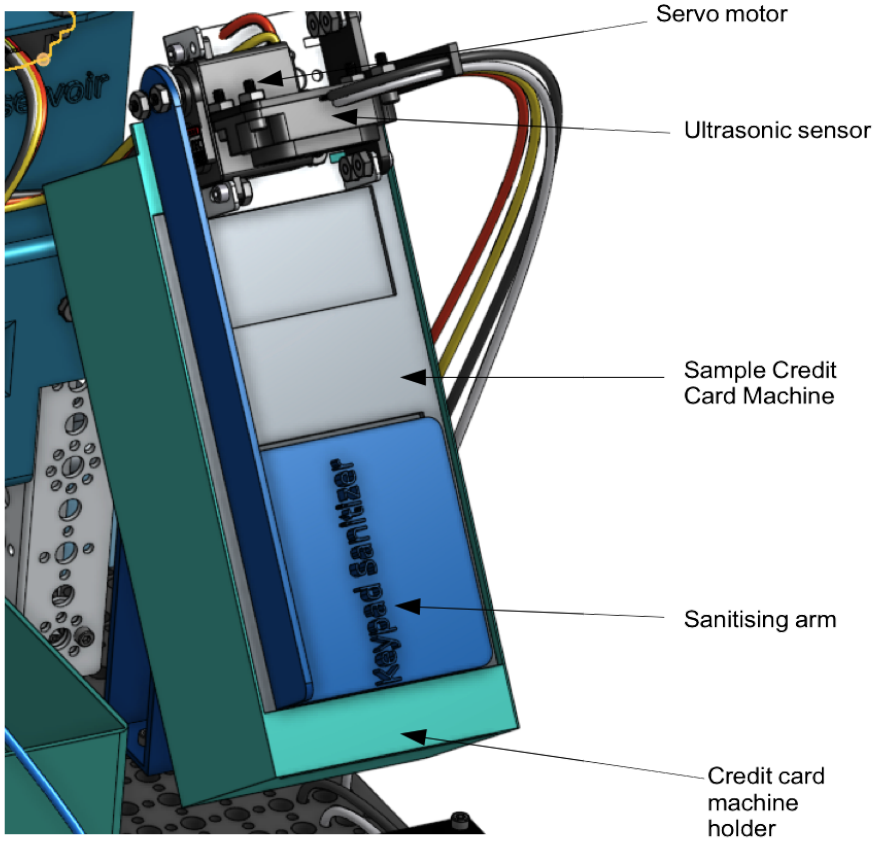
NOTE: The table sanitisation mechanism sanitises the tracks and whole table every single time it moves.

Mechanisms functions

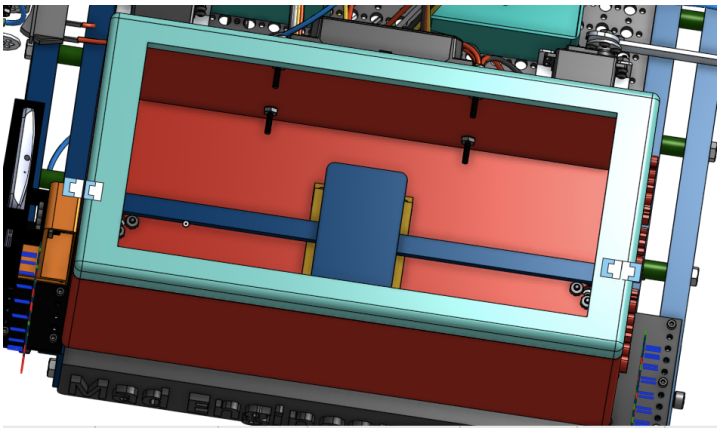


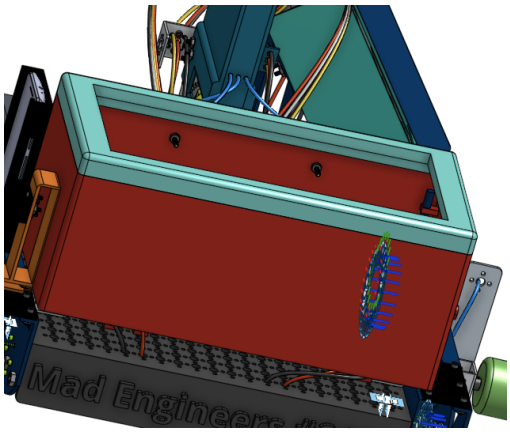
CASH AND COINS SANITISATION MECHANISM (INSIDE BLUE COVER)

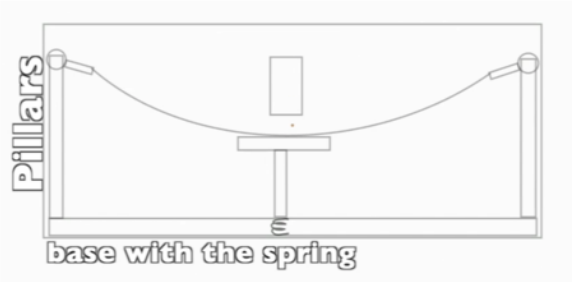
CREDIT CARD MACHINE HOLDER– holds portable credit card machine of standard size. Servo mechanism with moving arm sanitises the keypad regularly.



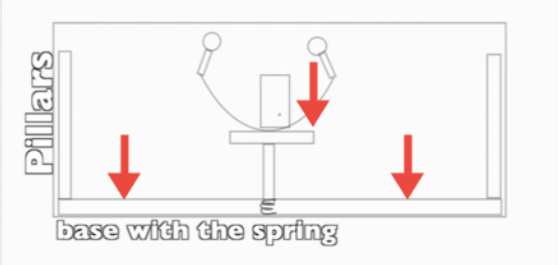
Packaging mechanism



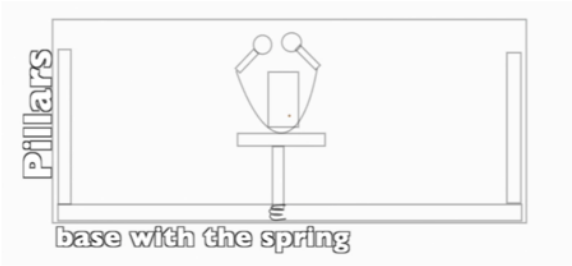




Original position



Downward application of pressure lowers spring and pillars, thereby releasing elastic bag



Elastic wraps around product

WORKS WITH ALL SHAPES AND SIZES OF PRODUCTS

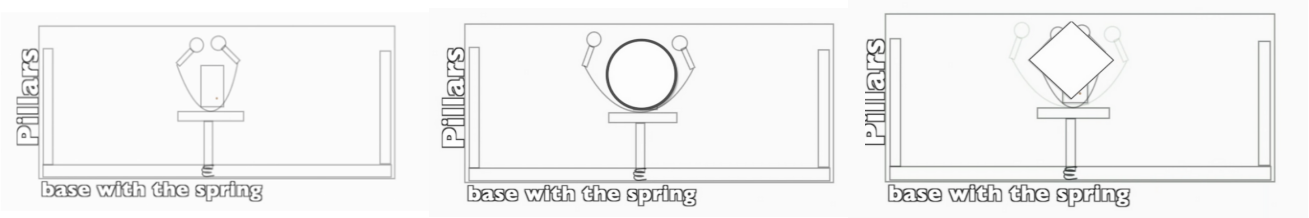


Table sanitisation mechanism

